Game 1 Catch the Butterfly

Before you start

These are teacher-designed Scratch 3 instructions. Work through the core game first, then test it with a partner. Save your project in the class folder.

Build the game

Choose a butterfly sprite and a garden backdrop. Remove the cat if it is not needed.

Add the two scripts below to the butterfly. The first script starts the game; the second runs when the player clicks the butterfly.

```
when green flag clicked
show
go to random position

when this sprite clicked
play sound [pop] until done
go to random position
```

Choose a short sound available in your project. If the sprite is too small to click comfortably, increase its size.

Test and personalise

Click the green flag, then click the butterfly several times. Each click should play a sound and move it. Test a new game with the green flag.

Add a short instruction with a say block at the start. Check that the speech does not hide the butterfly for too long. Choose a different backdrop or sound.

Optional teacher-supported score: create a Score variable for all sprites. Set Score to 0 in the green-flag script and change Score by 1 in the click script.

Find and fix

If a click does nothing, check that the script belongs to the butterfly and starts with when this sprite clicked. If it disappears, add show at the start.

Explain

What starts each script? Show one improvement you tested.

Sources and related skills

Related book pages are included after this guide. They teach supporting skills; this game is a separate classroom project.

Official Scratch project ideas: https://scratch.mit.edu/ideas

Game 2 Find the Treasure

Before you start

These are teacher-designed Scratch 3 instructions. Work through the core game first, then test it with a partner. Save your project in the class folder.

Set up a simple route

Choose a player sprite and a treasure sprite. Draw a clear route on the backdrop with a start and finish. Keep the first route wide and short.

Select the player. Put it at the start, then add a reset script using its starting x and y values. For example:

```
when green flag clicked
go to x: (-180) y: (-100)

when [right arrow] key pressed
change x by (10)

when [left arrow] key pressed
change x by (-10)

when [up arrow] key pressed
change y by (10)

when [down arrow] key pressed
change y by (-10)
```

Add a finish response

Place the treasure at the finish. In the player's green-flag script, after the reset, add:

```
forever
  if <touching [Treasure] ?> then
    say [You found it!] for (2) seconds
    stop [this script]
  end
end
```

Select the actual name of your treasure sprite in the touching block.

Test and improve

Use the arrow keys to reach the treasure. Press the green flag to restart from the same place. Ask a partner to follow the route.

The first version uses an on-screen route: it does not stop a player crossing a wall. Follow the route honestly. Collision walls are an optional teacher-led extension.

If movement is too large, reduce 10 to 5. Explain which blocks move sideways and which move up or down.

Sources and related skills

Related book pages are included after this guide. They teach supporting skills; this game is a separate classroom project.

Official Scratch project ideas: https://scratch.mit.edu/ideas

Game 3 Catch the Fruit

Before you start

These are teacher-designed Scratch 3 instructions. Work through the core game first, then test it with a partner. Save your project in the class folder.

Build or open the starter

Choose a basket and one fruit sprite. Position the basket near the bottom. Use a simple backdrop. The teacher may provide the falling-fruit script as a starter.

Select the basket and add:

```
when green flag clicked
go to x: (0) y: (-140)

when [right arrow] key pressed
change x by (15)

when [left arrow] key pressed
change x by (-15)
```

Keep the basket inside the stage during testing. Optional teacher support: limit x to -200 through 200.

Make the fruit fall and reset

Select the fruit. Build this script, using the basket's actual sprite name:

```
when green flag clicked
go to x: (pick random (-200) to (200)) y: (160)
forever
  change y by (-4)
  if <touching [Basket] ?> then
    play sound [pop] until done
    go to x: (pick random (-200) to (200)) y: (160)
  end
  if <(y position) < (-170)> then
    go to x: (pick random (-200) to (200)) y: (160)
  end
  wait (0.03) seconds
end
```

Test and improve

Test a catch and a miss. Both should send the fruit back to the top so play continues.

If fruit falls too fast, change -4 to -2. If a catch makes no sound, check the touching block and that both sprites can meet.

Optional score: reset Score to 0 when the game starts and add 1 only inside the touching-basket branch.

Explain

Point to the repeated falling action, the catch check and the reset after a miss.

Sources and related skills

Related book pages are included after this guide. They teach supporting skills; this game is a separate classroom project.

Official Scratch project ideas: https://scratch.mit.edu/ideas

Game 4 Animal Quiz

Before you start

These are teacher-designed Scratch 3 instructions. Work through the core game first, then test it with a partner. Save your project in the class folder.

Ask one question

Choose an animal sprite and a simple backdrop. Begin with one question and a typed answer. Build this script on the sprite:

```
when green flag clicked
ask [Which animal says meow?] and wait
if <(answer) = [cat]> then
  say [Correct!] for (2) seconds
else
  say [A cat says meow. Try the next one.] for (2) seconds
end
```

The answer reporter comes from Sensing. The equals operator compares the answer with the expected word.

Add two more questions

Duplicate the ask-and-check section below the first question, inside the same green-flag script. Change both the question and the expected answer.

Try: Which animal says woof? Expected answer: dog. Which animal has a trunk? Expected answer: elephant.

Keep each question short. Agree acceptable words in advance; the first version accepts the specified word, not every possible sentence.

Test both branches

Enter the correct answer for each question, then restart and deliberately enter an incorrect answer. Check that feedback matches the question.

If an answer is wrongly rejected, check spelling and accidental spaces in the expected answer. If the wrong feedback appears, check the duplicated question's answer block.

Personalise and explain

Change the animal, backdrop or questions after the first quiz works. Keep the first quiz to two or three questions.

Show the block that asks a question, the answer reporter and the two feedback paths.

Sources and related skills

Related book pages are included after this guide. They teach supporting skills; this game is a separate classroom project.

Official Scratch project ideas: https://scratch.mit.edu/ideas

What is AI

Teacher reference

Original classroom activities for Year 2. Use teacher-led demonstrations, prepared predictions and non-personal pictures.

Know the difference

Some AI systems use patterns learned from examples to make predictions, such as guessing what an image shows. A simple timer follows fixed instructions. A device being called smart does not tell us whether it uses AI.

Try it

Give pairs six picture cards: timer, calculator, a basic Scratch movement script, a described picture-recognition system, a speech-recognition system and an unfamiliar gadget.

Sort the cards into uses AI in this example, fixed instructions, and need more information. Ask pupils to explain one choice. Do not guess about an unknown gadget.

Check

Ask for one AI example and one program that follows fixed instructions. Can the pupil explain the difference in simple words?

Teacher sources

AI4K12 grade-band progression charts (K-2): https://ai4k12.org/gradeband-progression-charts/

UNICEF Guidance on AI and Children: https://www.unicef.org/innocenti/reports/policy-guidance-ai-children

Learning from examples and checking mistakes

Teacher reference

Original classroom activities for Year 2. Use teacher-led demonstrations, prepared predictions and non-personal pictures.

Lesson 1 Examples and labels

An AI classifier can use labelled examples to learn patterns. This classroom card activity is a model of the idea; pupils sorting cards are not a real AI system.

Prepare several fish pictures and several non-fish pictures. Include different shapes, colours and sizes. Label each example and ask what features help tell them apart.

Show a new picture and ask pupils to predict a label. Compare using varied examples with repeatedly showing the same picture.

Lesson 2 A prediction can be wrong

Prepare a results sheet with some correct and incorrect labels. Include a fish-shaped toy labelled real fish.

Pupils mark which predictions need checking. Compare the picture, its correct label and the system's guess. Explain why a person should check unfamiliar examples.

Do not describe a confident answer as proof. Ask what extra examples or information might help.

Check

Can the pupil explain why varied examples help and identify one incorrect prediction? Accept an oral explanation or drawing.

Teacher sources

AI4K12 grade-band progression charts (K-2): https://ai4k12.org/gradeband-progression-charts/

UNICEF Guidance on AI and Children: https://www.unicef.org/innocenti/reports/policy-guidance-ai-children

Using AI safely and designing a helper

Teacher reference

Original classroom activities for Year 2. Use teacher-led demonstrations, prepared predictions and non-personal pictures.

Lesson 1 Ask an adult and protect information

Use fictional cards showing an app asking for a name, a photo, a voice recording, a password or a home address. Do not collect pupils' real details.

Pupils practise: stop, do not submit the information, and ask a trusted adult. Teacher demonstrations use non-personal pictures and school-approved tools.

Draw a stop-and-ask reminder. Explain that an unfamiliar AI tool needs adult help before use.

Lesson 2 Design a useful helper

Draw an AI helper that sorts pictures of classroom supplies. Name the categories and show several different examples it could learn from.

Add a difficult picture and explain one mistake the helper might make. Draw a person checking an uncertain result.

Explain one use, one limitation and one privacy or checking rule. This is a design activity; no pupil account or live AI submission is needed.

Check

Can the pupil name information to protect, demonstrate asking an adult, and explain a useful helper's limit?

Teacher sources

AI4K12 grade-band progression charts (K-2): https://ai4k12.org/gradeband-progression-charts/

UNICEF Guidance on AI and Children: https://www.unicef.org/innocenti/reports/policy-guidance-ai-children

4.1 Choose a sprite

In this lesson

You will learn:

- how to choose a sprite for your program.

Spiral back



Last year you used a simple program. It controlled a sprite. Now you will write a program to control a sprite.

In Unit 3 you played the Frog Hop Game. In this unit you will make a simple version of the game. You can choose any sprite you like. It does not have to be a frog.

You will make a Scratch program. Start on the Scratch website.

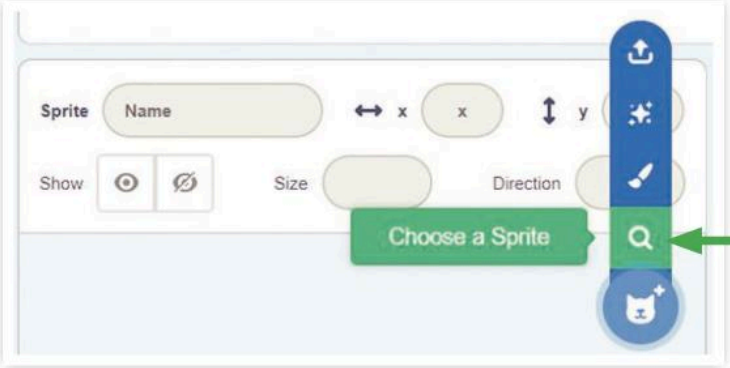
Choose a sprite

Look at the bottom right. This part of the screen shows the sprites.



Sprite 1 is a cat. Delete this sprite by clicking on the cross.

You will choose a new sprite.



Click here to see the other sprites.

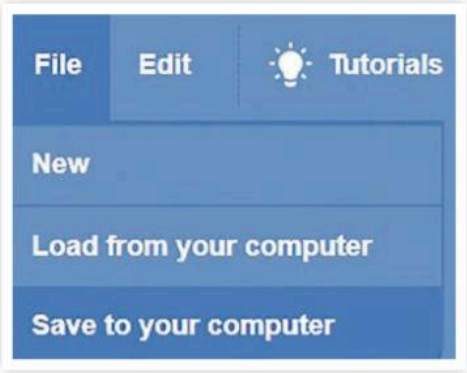
You will see pictures of sprites.



Click on the one you choose.

Save program file

Now you must save your work. You will save it on your computer. Open the File menu and choose 'Save to your computer'. Then type a name for your file.





Activity

Use the skills you learned to:

- pick a sprite
- save your file.

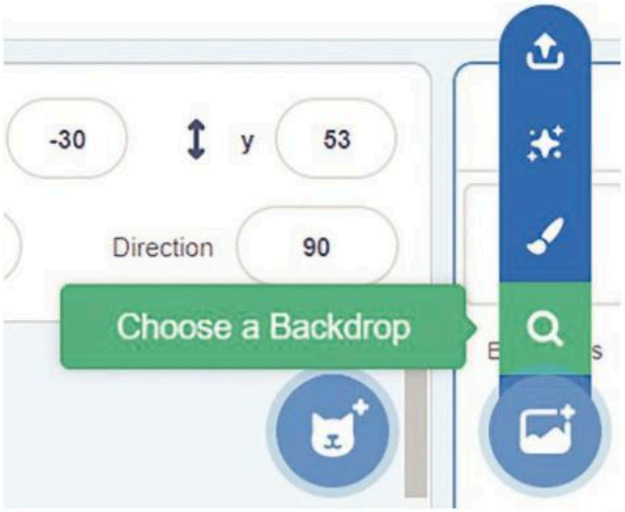


Draw the sprite you chose. Imagine a computer game with your sprite in it. Write down your idea.



Extra challenge

Add a backdrop to your program.



4.2 Command your sprite

In this lesson

You will learn:

➔ how to add commands to your program.

Load program file

Last lesson you saved your work as a file. Now you will load the file. Load means get a file from storage.



Open the file menu. Click on 'load from your computer'. Choose the file you made last time.

Coloured dots and blocks

You will make a Scratch program. A program is made of **commands**. Commands tell the computer what to do.

Scratch commands are blocks. You fit the blocks together. That makes the program.

Different types of blocks are different colours.

- 'Motion' blocks are blue.
- 'Sound' blocks are pink.

Look for the coloured dots on the left of the screen. Choose a colour to help you pick the blocks you need.

Actions

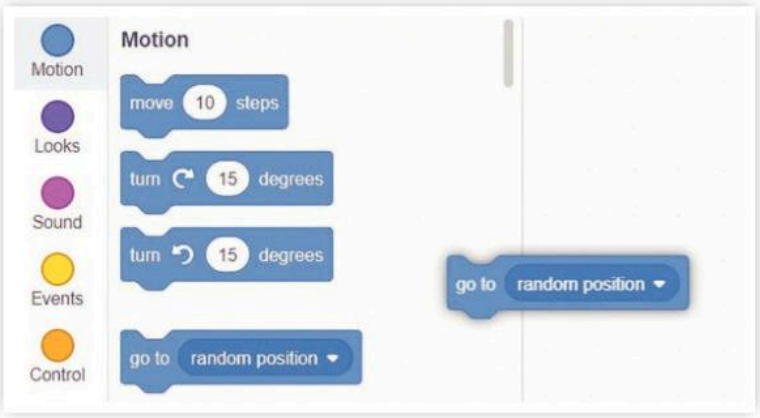
Choose a block. It will make your sprite move.

Click on the blue dot. You will see 'Motion' blocks.

Find the block that says 'go to random position'.

Drag this block to the middle of the screen.

Click on the block. The sprite will move.



Sounds

Choose a block. It will play a sound. Click on the pink dot. You will see 'Sound' blocks. Choose the 'play sound' block.

Now you have two blocks. Fit the blocks together.

Click on the two blocks. What happens?





Activity

Make the two-block program shown on this page.



Extra challenge

Add any other block to the program. See what happens when you run the program.

Think again

Say what blocks you used. Say what each block did.

4.3 A start event

In this lesson

You will learn:

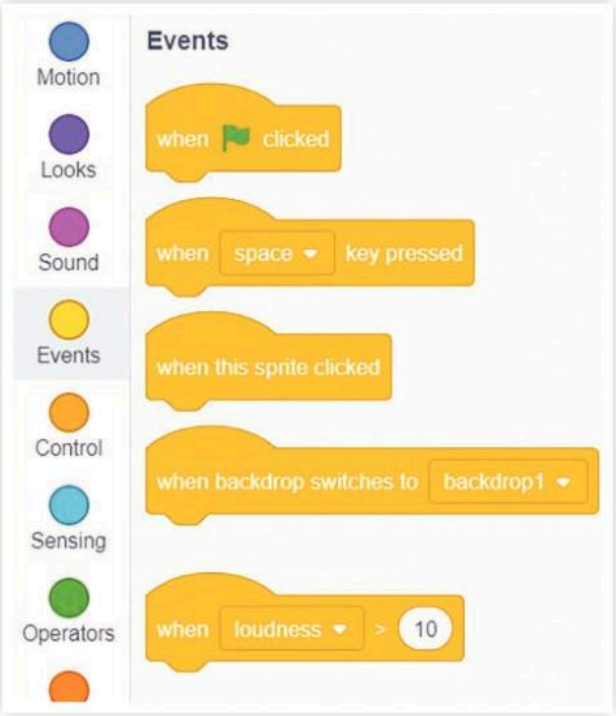
- to choose an event that starts the program.

Events

An **event** means anything that happens. In Scratch 'Events' are things the **user** does. The user is the person who uses the program. Now you will choose the 'Event' that lets the user start your program.

Click on the yellow dot that says 'Events'.

What 'Event' blocks can you see? What does each one say?

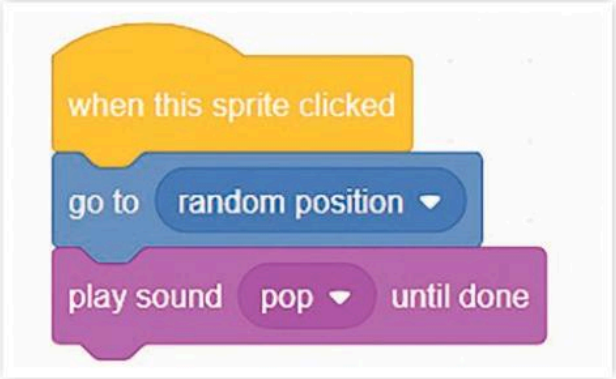


Choose an event

To choose an 'Event' drag the block to the program area.

Find the 'Event' block that says 'when this sprite clicked'.

Fit the program blocks onto the 'Event' block. Now the event will start the program.



Change colour

Now you will make the sprite change colour. Click on the purple dot that says 'Looks'. These command blocks change the look of the sprite.

Find the block that says 'change color effect by…'.

Drag this block to the script area and add it to the program.





Activity

Change your program.

- Add the 'Event' block which says 'when this sprite clicked'.
- Add the 'Looks' block which changes the colour of the sprite.

Click the sprite to make it move.

Save the file.



Extra challenge

Choose a different start event. Change the program to the new event.

Think again

Draw the 'Event' block you used. Say what it does.

If you used more than one, draw them all.

4.4 Forever loop

In this lesson

You will learn:

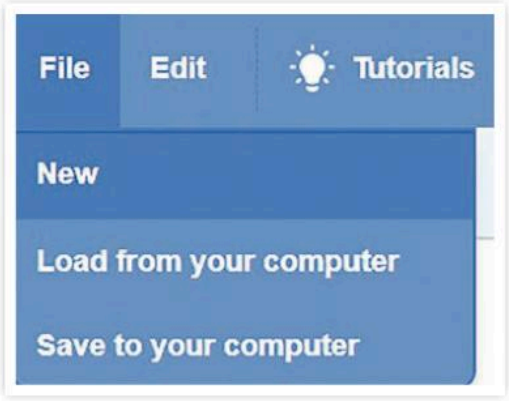
- how to change the program so the sprite moves 'forever'.

Last lesson you wrote a program. The program started when you clicked on the sprite. The sprite moved to a new place. The sprite made a sound. These things happened one time.

Now you will make a new program. In this program the sprite will move over and over again. The action will repeat.

New program

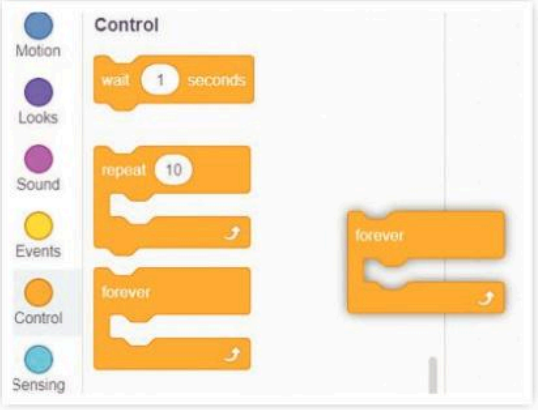
Open the file menu. Click on 'New' to make a new program.



Control blocks

Click on the orange button. You will see the orange 'Control' blocks. Find the block that says 'forever'. This block is called the **forever loop**. Drag it to the middle of the screen.

A forever loop is a way to control the program. Commands inside the forever loop will repeat.



Add commands

Here are the commands you want to repeat:

- Make the sound 'pop'.
- Go to a random position.

Put these blocks inside the forever loop. They are the same blocks you used in the last program.

Wait

Add one more block inside the forever loop. You will find it with the orange control blocks. It says 'wait 1 seconds'. This will make the sprite wait for 1 second between jumps.

The completed program looks like this.





Activity

Start a new program. Choose a sprite and a background.

Make the program shown on this page. Run the program to see what happens.

Think again

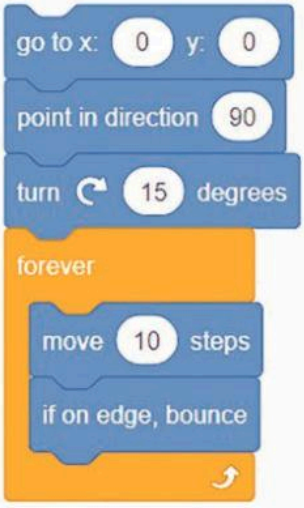
Say what happens when you put blocks inside a forever loop.



Extra challenge

Here is a new program that uses the forever loop.

- Make this program.
- Add a start 'Event' to the program.
- Change the numbers in the program and see what happens.



4.5 Plan and make

In this lesson

You will learn:

➔ how to make a new program to match a plan.

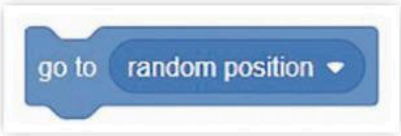
Plan

Here is the plan.

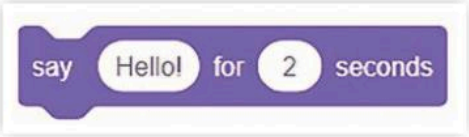
- The sprite will jump to a random place.
- The sprite will say "Hello!"
- The sprite will keep moving towards the mouse pointer.

First commands

The sprite will start in a **random** place. Random means you do not know where it will go.



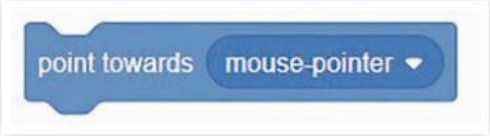
The sprite will say "Hello!"



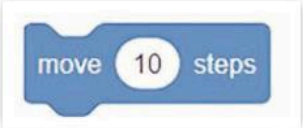
Find these blocks and fit them together.

Repeated commands

Here is the block that makes the sprite point towards the mouse pointer.



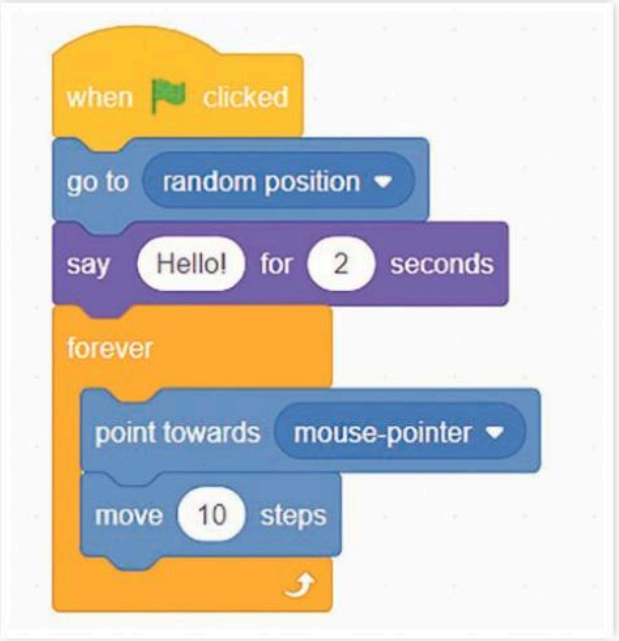
Here is the block to move 10 steps.



These two blocks will go inside a forever loop.

Completed program

Here is the program with all the pieces in place. It has a start 'Event' – the green flag.





Activity

Start a new program. Choose a sprite that looks like a fish. Choose a background that looks like underwater.

Make the program shown on this page. The program will make the fish swim about underwater.

Think again

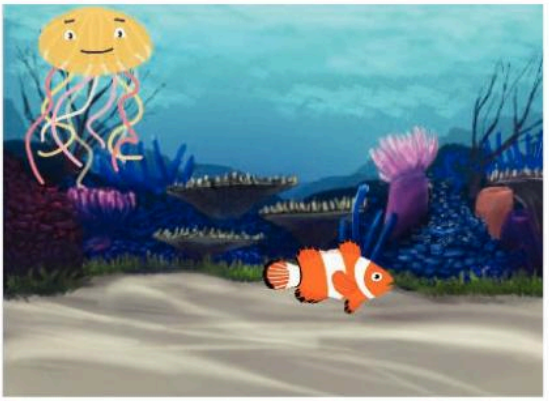
Think of one change you could make to the program you made. What would happen if you made this change?



Extra challenge

Add a jellyfish sprite to the underwater scene. Make a program for this sprite.

- Use the same blocks as before.
- Take out the block that says 'follow the mouse pointer'.
- Add a block that says 'if on edge bounce'.
- Make the number of steps 2 instead of 10.



4.6 Find and fix

In this lesson

You will learn:

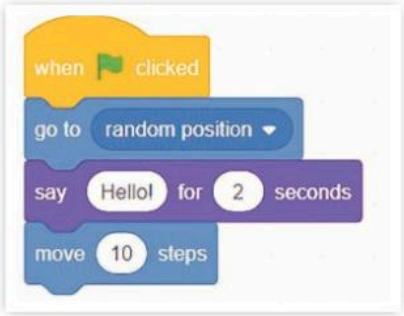
- how to make changes to a program
- how to find and fix errors.

Commands to match actions

A student wanted to make a program. They decided the sprite would:

- jump to a random place
- say "Hello!"
- move around the screen forever.

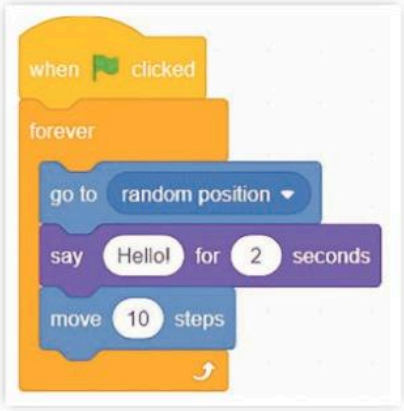
The student found blocks to match the actions. Here is the program made of those blocks.



The student ran the program. The program did not work the way they wanted. The sprite moved just once, then stopped.

Using a loop

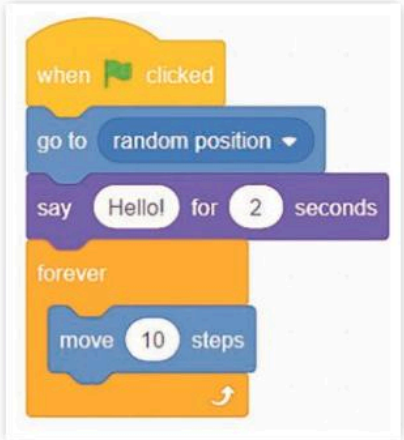
The student decided to use a 'forever loop'. They put all the command blocks inside the loop. Here is the program they made.



The student ran the program. The program did not work the way they wanted. Too many blocks are inside the loop. The sprite keeps jumping to a new place. The sprite keeps saying "Hello!"

Inside and outside the loop

Some commands should go before the loop. Those are the commands that must happen only once.



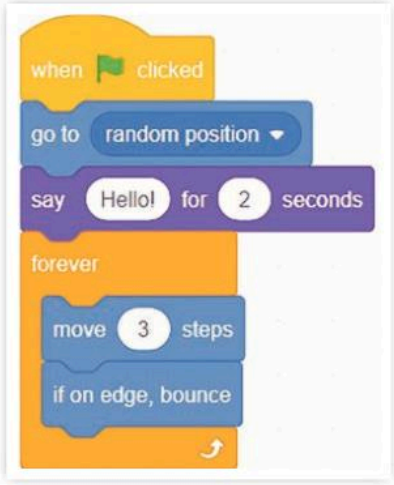
Some commands should be put inside the loop. Those are the commands that will be repeated.

Look at the final picture on the previous page. You will see the program with all the commands in the right place.

Add some bounce

The sprite moves until it is almost off the screen and then stops. You must add a block that tells the sprite to 'bounce' if it hits the side.

The finished program is shown here.





Activity

Using the ideas on this page make a program that makes a butterfly move about the screen.



Extra challenge

Look in the program you made to find the block that says 'move 10 steps'.

Change the number. Run the program and see what is different.

Explore what happens when you use different numbers.



Explore more

Make a new program. Use a sprite that looks like a ball. Make it bounce around the screen.

Add different types of ball. Set them to different speeds.

